



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

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- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer:

This program keeps track of incident tickets created when bots fail. Each ticket has an Incident ID, a bot name, and a short description. The tickets are stored in a Python list, and a menu lets the user add a ticket, display all tickets, search by ID, remove a resolved ticket, and count the tickets that are still active.

- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident:

```
"tickets.append([input("Incident ID: "), input("Bot: "), input("Description: ")])"
```

The program asks for the Incident ID, bot, and description, puts them in a list, and uses append() to add that ticket to the end of the tickets list.

Display Incidents:

```
for t in tickets:  
    print(t[0], "|", t[1], "|", t[2])
```

The loop goes through every ticket in the list and prints its Incident ID, bot, and description.

Search Incident:

```
for t in tickets:  
    if t[0] == id:  
        print("Found:", t[0], "|", t[1], "|", t[2])  
        break  
else:  
    print("Not found.")
```

The loop compares each ticket's Incident ID (t[0]) to the ID entered by the user. If one matches, the ticket is printed and the loop stops. If no ticket matches, the else prints "Not found."



Remove Resolved Incident:

```
if tickets[i][0] == id:  
    tickets.pop(i)  
    print("Removed.")  
    break
```

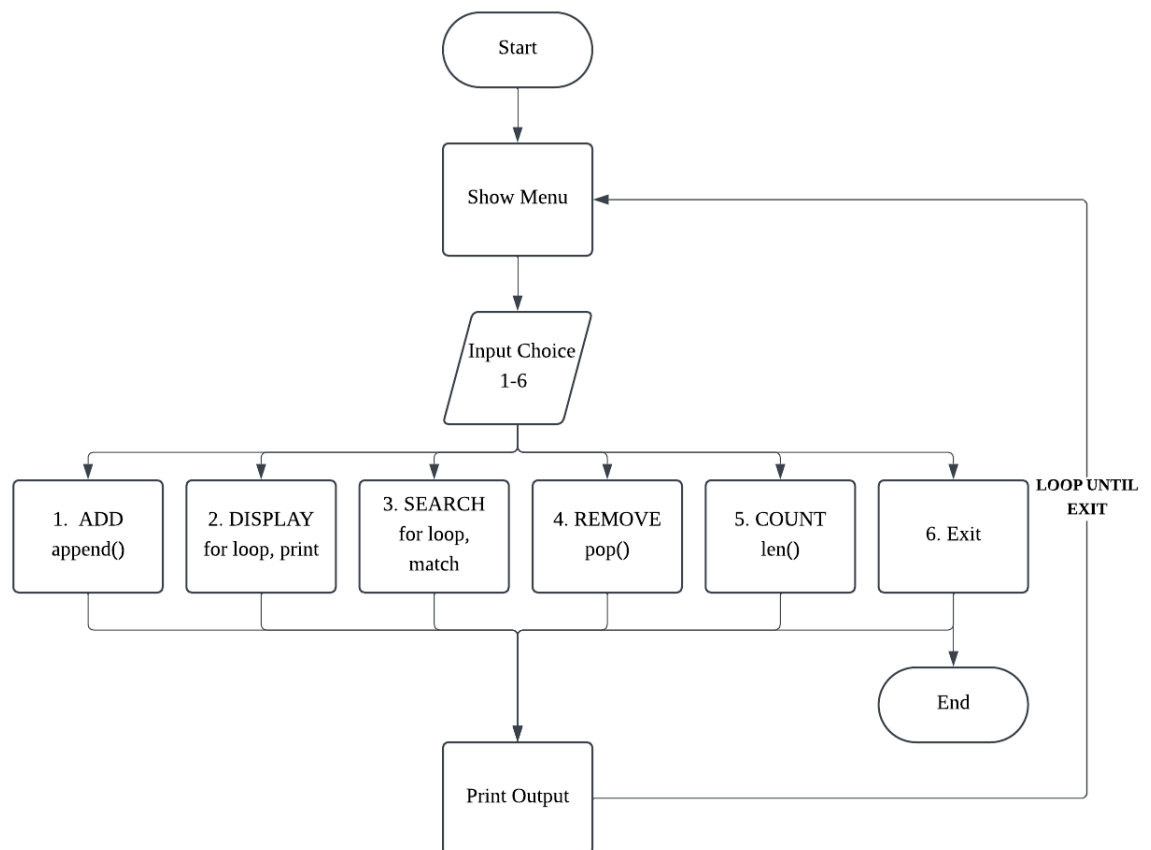
The loop finds the ticket with the matching Incident ID, and remove() deletes it from the list. The loop then stops, so the ticket is no longer active.

Count Active Incidents:

```
print("Active tickets:", len(tickets))
```

The len() function returns the number of tickets left in the list, which is the total number of active incidents.

III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.





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- IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

This activity showed how a linear data structure can be used to store and manage a collection of records. The program uses a Python list to hold the incident tickets, with each ticket stored in order of arrival. Adding a ticket uses `append()` to place it at the end of the list. Displaying and searching use a loop that visits each ticket one by one, from the first element to the last. Removing a resolved ticket deletes the matching element, and `len()` gives the current number of active tickets. The activity also showed how a menu inside a loop keeps the program running until the user chooses to exit. Overall, it demonstrated that a linear structure is a simple and effective way to manage a small set of records, though searching gets slower as the list grows because every element may need to be checked.